Devices of Calculation

The word “calculation” derives from the Latin “calculi”, the word for a small stone. For keeping track of sheep and cattle, using small stones is quite a good method, but once numbers get big, and once we want to do operations such as addition or multiplication, the small stone system will have to be updated.

The abacus cleverly gets around this problem by assigning some “stones” (usually called beads) to be worth more than others. Typically, beads in the top row are worth five units and beads in the bottom row are worth 1 unit.

Unfortunately for the abacus, the Universe doesn’t move along in discrete integer steps, for the most part, the Universe is continuous. Clever mathematicians found methods to preserve the abacus, but ultimately something new was needed, the slide rule. The laws of logarithms paved the way for sliding scales to become the gold standard in calculation technology, up until very recently. The calculations for the Moon landings were done on devices like these.

More recently, the hand-held digital calculator has taken over as the go-to device. They are much more precise and allow more complex operations to be done. Many today have forgotten how recent these helpful devices are. Up until 1990, the use of digital calculators was prohibited in the Leaving Certificate exam.

`Traditional Chinese: "Abacus" (calculating tray).`

Devices of Callan

Reverend Nicholas Joseph Callan was born on the 22nd of December 1799 in Dundalk, Co. Louth. He first enrolled in Maynooth College in 1816, where he studied divinity and natural philosophy; today known as physics. In 1826, Callan received his doctorate in divinity in Rome, after studying there for three years. While in Italy, he became inspired by those such as Alessandro Volta and Luigi Galvani, two notable physicists who specialised in electromagnetism.

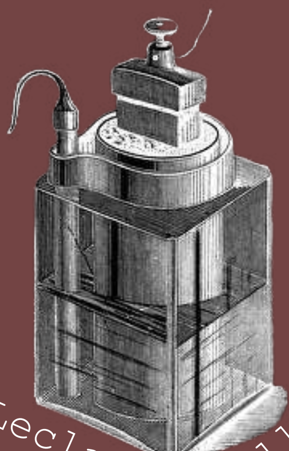
Callan was a brilliant experimentalist, eventually inventing the induction coil, building upon the theory of Micheal Faraday. In his attempts to measure the current created by his batteries, Callan commissioned a local blacksmith to create a massive horseshoe, upon which Callan would coil wire, creating an electromagnet. This great electromagnet was then used to pick up weights of iron, the weight lifted being proportional to the current though the wires. Callan's great electromagnet was capable of lifting over one metric tonne of iron when connected to his 577-cell battery.

In 1837, Callan developed the largest induction coil he would ever create. He used a clockwork-repeater to interrupt the current at a frequency of 20Hz, creating a sparks of 15" (~38 cm), the largest artificial bolt of electricity seen up until that point. Today, it is estimated that these arcs must have jumped across a voltage of around 600,000 volts.

Devices of Current

Generating the current required to operate an induction coil is a battle in and of itself. Today, many are familiar with the lemon battery; copper and zinc plates separated by a lemon-juice-electrolyte creates a voltage between the plates. This was the basis of the “voltaic pile”, the first designed battery ever constructed. This device is effectively many lemon batteries strung together in series, instead of lemon juice it uses cloth soaked in acidified water, however.

One of the many problems with the voltaic pile, was that as time went on its current output decreased significantly.
Ultimately, the “constant battery” was invented, a battery whose current output was stable up until a critical point.
The Leclanché cell and the Maynooth battery were some of the first constant batteries to be created.



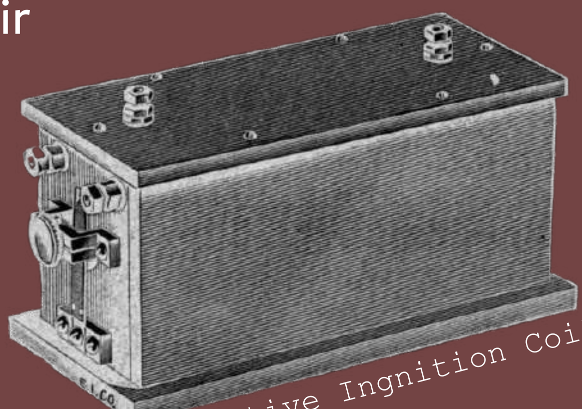
Leclanché Cell

The Maynooth battery, designed by Nicholas Callan, was one of the first batteries invented which used affordable materials. The battery mostly consists of a cast-iron casing and a zinc plate, as well as a single acidic solution. Most batteries at the time had to use expensive elements such as platinum.Callan also constructed the world’s largest battery of the age, joining 577 cells together into one large battery, which used over 100 litres of acid.

Devices of INDUCTION

In 1931, Michael Faraday discovered that a changing current in one coil of wire would induce a current in a second, adjacent loop of wire. This concept was then expanded upon by Nicholas Callan, who realised that the voltage on the second coil could be stepped-up or stepped-down by changing the ratio of turns in each coil

In a car engine, induction coils are used to generate sparks within the cylinder – which detonates a fuel-air mixture and powers the engine. When electricity enters your house from overhead cables, it needs to be stepped-down from thousands of volts to only a couple hundred. An inductive device, the transformer, is used for this purpose.



Automotive Ingnition Coil

$$I_2 = \frac{A_2 - I_1 \frac{dV}{dt}}{R_2}$$

Maxwell's Oringal Induction Expression

Callan used a spark gap to measure the output voltage of the secondary coil, in one of his medium sized coils, he could generate arcs of up to 5 inches, which corresponds to a voltage of around 200,000 volts!

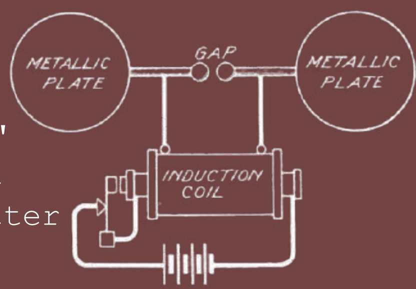
Callan’s induction coil inspired further improvements by William Sturgeon, enabling the developments of critical fields, such as wireless communication and computation.

Devices of Marconi

Guglielmo Marconi was born in Bologna, Italy in 1874. His mother, Annie Jameson was the granddaughter of John Jameson, the founder the famous Irish whisky company bearing his name. His mother's heritage gave him a life-long connection to Great Britain and Ireland (the two of which were unified at the time).

Marconi is most famous for his creation of the first radio-wave wireless telegraphy system and is today credited as the inventor of the radio. From a young age, Marconi was interested in science, particularly in transmitting signals without using wires, as was the case for the telegraphic systems of the day.

Hertz' First Transmitter



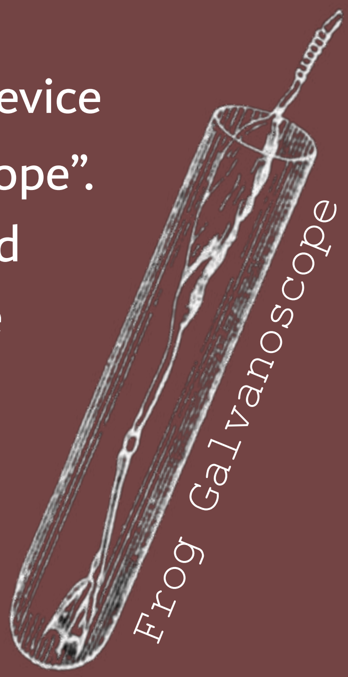
Marconi did most of his experimenting in his family's attic with the assistance of his butler. Eventually, Marconi developed a method of transmitting and receiving radio waves, up to about half a mile, but this was not far enough for his device to be commercially successful.

In 1895, Marconi made a breakthrough, realising that grounding his transmitter and receiver increased the distance over which the waves could be transmitted. Eventually, Marconi patented his invention in London, and radio as we know it today was born. By the end of Marconi's life, messages were being transmitted across the Atlantic, the first of which was sent from Clifden, Co. Galway by Marconi and his company.

Devices of Measurement

Once we create electricity, the task becomes figuring out just how much we've created. Until the mid-19th century, most physicists had to make subjective measurements, usually observing the maximum length of electrical arcs produced by a voltage. A rough approximation is that every centimetre of electrical arc requires around ten thousand volts.

In the late 18th century, Luigi Galvani invented a device for measuring voltage, called the "Frog Galvanoscope". As the name suggests, the device used the severed leg of a frog to indicate to the user the magnitude of voltage across the leg.



Ultimately, in 1820, Hans Christian Ørsted discovered that electrical currents will deflect compass needles placed near them, and that electric currents must produce magnetic fields. Johann Schweigger and André-Marie Ampère invented what would become the galvanometer, named after the earlier amphibian-based detector invented by Galvani.

The difference between galvanoscopes and galvanometers is that the galvanoscope gives an indication of the voltage between two electrodes, but the galvanometer has a scale from which this voltage can be objectively measured.

Devices of Novelty & Magic

On the 21st of April 1820, magic entered the world through a lecture theatre in Copenhagen. While setting up his lecture, Hans Christian Ørsted noticed that upon turning on an electric current by connecting a wire to a battery, a nearby compass needle deflected away from north. The deflection was slight enough that no students noticed, but Ørsted caught it – the Universe was trying to pull a fast one.

The unification of electricity and magnetism came as a shock to everyone who could understand what those words meant. Two separate phenomena, in an instant, were coupled together in a dance for evermore.

Those who formulated the laws of electromagnetism could have reasonably been considered magicians at the time. In order to bring electromagnetism into the reality of the 19th century, many devices were created to demonstrate the reality of the situation, to show that laws still had to be followed.

To attract interest, many of these devices included an element of fun, electric cannons and music machines just as examples. Today we take electromagnetism for granted, it's an everyday affair, but these machines give us a view into the beginnings of our world. For a brief period in the mid-19th century, magic really existed – and it flowed through copper wires.

Maxwell's Equations

$$\nabla \cdot \mathbf{E} = \frac{\rho}{\epsilon_0} \qquad \nabla \cdot \mathbf{B} = 0$$

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t} \qquad \nabla \times \mathbf{B} = \mu_0 \mathbf{J} + \frac{1}{c^2}\frac{\partial \mathbf{E}}{\partial t}$$

The National Science Museum

is opening for the summer

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Thursday 13:00-19:00
Friday 12:00-18:00
Saturday 11:00-16:00

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The Electric Field

"The sciences are monuments devoted to the public good; each citizen owes to them a tribute proportional to his talents. While the great men, carried to the summit of the edifice, draw and put up the higher floors, the ordinary artists scattered in the lower floors, or hidden in the obscurity of the foundations, must only seek to improve what cleverer hands have created."

Charles-Augustin de Coulomb

$$\vec{E} = -\nabla\phi - \frac{\partial\vec{A}}{\partial t}$$

The electric field describes the force applied to a positive test charge at every point in space. Positive charges create diverging electric flux, while negative charges create converging flux. Gauss discovered that the amount of flux flowing through a region is proportional to the amount of charge in that region:

$$\oiint_S \vec{E}\cdot d\vec{S} = \frac{Q}{\epsilon_0}$$

The electric field is the driving force behind most electric currents. It governs everything from the operation of toasters to allowing chemistry to exist at all. Much of the experimental work relating to the electric field was conducted in the 18^{th} and early 19^{th} centuries by scientists such as Michael Faraday and Nicholas Callan, the elucidating theory arriving soon after.

The experimentation of Coulomb revealed the electric field to decrease as the square of the distance from a charge, drawing parallels between the electric and gravitational field - suggesting the prevalence of inverse-square laws in nature was more exaggerated than previously expected.